

Finding the Equation of a Perpendicular Line

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Question:

A straight line L is perpendicular to the line $5x - y + 1 = 0$. The area of the triangle formed by L and the coordinate axes is 5. Find the equation of the line L .

Solution:

Let the given line be L_1 and the required perpendicular line be L . The equation of the given line L_1 is $5x - y = -1$. Using the matrix form $\mathbf{n}^\top \mathbf{x} = c$, its parameters are:

$$\mathbf{n}_1 = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad c_1 = -1 \quad (0.1)$$

Let the required line L have the equation $\mathbf{n}_2^\top \mathbf{x} = c_2$. Since the lines are perpendicular, the dot product of their normal vectors must be zero, so $\mathbf{n}_1^\top \mathbf{n}_2 = 0$.

Let's assume the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ m \end{pmatrix}$. We can solve for m :

$$\begin{aligned} \mathbf{n}_1^\top \mathbf{n}_2 &= \begin{pmatrix} 5 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ m \end{pmatrix} = 5 - m = 0 \\ \implies m &= 5 \end{aligned} \quad (0.2)$$

Therefore, the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$, and its equation is $x + 5y = c_2$.

The line L forms a triangle with the coordinate axes. We find the intercepts. For the x -intercept, set $y = 0$, which gives $x = c_2$. The point is $(c_2, 0)$. For the y -intercept, set $x = 0$, which gives $5y = c_2$, so $y = c_2/5$. The point is $(0, c_2/5)$.

The area of this triangle is given as 5. Using the area formula:

$$\begin{aligned} \text{Area} &= \frac{1}{2} |(\text{x-intercept}) \cdot (\text{y-intercept})| = 5 \\ \frac{1}{2} |(c_2)(c_2/5)| &= 5 \\ \left| \frac{c_2^2}{10} \right| &= 5 \\ c_2^2 &= 50 \implies c_2 = \pm \sqrt{50} = \pm 5\sqrt{2} \end{aligned} \quad (0.3)$$

There are two possible lines that satisfy the given conditions. Substituting the values

of c_2 into the matrix equation $\mathbf{n}_2^\top \mathbf{x} = c_2$, the possible equations for line L are:

$$\begin{pmatrix} 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 5\sqrt{2} \quad (0.4)$$

$$\begin{pmatrix} 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -5\sqrt{2} \quad (0.5)$$